LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	24
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Need for cleaning the electrical connectors Use of heavy-duty cleaner, cleaning fluid for electrical, electronics and allied components.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Need for cleaning the electrical connectors Use of heavy-duty cleaner, cleaning fluid for electrical, electronics and allied components. Safety precautions & Practical on cleaning of connecters, terminals.
5. Specific Objectives	13.1.4.1 knowledge on Safety precautions & Practical on cleaning of connecters, terminals.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on cleaning of connectors, terminals using heavy-duty cleaner and cleaning fluid for electrical components.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Why are electrical connectors critical in avionics systems?
(b) What are the common causes of connector failure in aircraft?
(c) List three safety precautions required before handling cleaning fluids.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Need for cleaning electrical connectors • Types of contamination in avionics • Consequences of dirty connectors • Selection of cleaning agents • Properties of heavy-duty cleaner	L & Ds	PPT, IB	15 min
(b)	• Safety precautions for cleaning fluids • Handling electrical components safely • Proper disposal of waste fluids • Personal protective equipment usage • Ventilation requirements in workspace	L & Ds	PPT, PDF	20 min
(c)	• Practical cleaning of connectors • Application of cleaning fluid • Inspection of cleaned terminals • Drying and reassembly procedures • Verification of electrical continuity	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the primary reason for using heavy-duty cleaner on electrical connectors.
(b) Describe the correct procedure for applying cleaning fluid to a terminal.

12. Home Assignment and Follow Up	Prepare a brief report on the differences between cleaning fluids for general electronics and those specific to avionics, referencing the next lesson on motor operations.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Practical on operation on DC motor, Induction Motor (1Ø & 3 Ø), universal motor on AEMT.	Refer to the master syllabus reference list	LP 25

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min